KRISH SHAH

AI Systems Portfolio

Four builds that show what production-grade AI actually looks like.
Each project combines AI logic with real architecture decisions, measurable outcomes, and systems thinking.

AI Agents · Systems Engineer · Architect · IREB Certified

PROJECT INDEX

Four production AI systems

Phonx AI

Voice AI Platform

Real-time voice automation for insurance and call-heavy operations. Handles multi-turn conversations with sub-second response, compliance logging, and CRM sync.

<1s response · 90%+ accuracy · 60%+ call reduction

Enterprise VKG

Knowledge + Compliance

Hybrid semantic and graph retrieval for audit-ready regulatory validation across 7 governing bodies. 5,000+ documents indexed on-prem with zero cloud dependency.

5K+ docs · 98%+ accuracy · 7 regulators

Satellite ML

Poppy Detection

Geospatial ML pipeline that detects suspicious cultivation using custom CNN trained on spectral signatures, outputting GPS-precise GIS shapefiles for field operations.

0.87 F1 · 5 min per scene · GIS-ready

Avara Labs

Email Automation

Modular email automation for e-commerce support. Classifies intent, routes AI-safe vs human-escalation, generates brand-safe replies grounded in company policy and SOPs.

~80% automated · Multi-brand · GDPR compliant

PHONX AI

Voice AI Platform for Insurance & Call-Heavy Operations

Industry: Insurance | Role: AI Product Engineer & Head of Dev

<1s
avg response time

90%+
intent accuracy

60%+
call reduction

A microservice-based voice automation system spanning telephony, speech services, LLM reasoning, compliance enforcement, and CRM synchronisation. Built for sub-second, multi-turn conversations. A proprietary insurance knowledge engine handles domain-specific terminology, policy nuances, and regulatory boundaries. The system transforms phone operations from a human bottleneck into a scalable infrastructure layer.

Telephony Gateway
SIP call ingestion & intelligent routing

STT / TTS Services
Sub-100ms audio conversion

Conversation Manager
LLM reasoning, multi-turn context

Knowledge Engine
Proprietary insurance retrieval system

Compliance Module
Tamper-evident logs for every decision

CRM Connector
Policyholder data sync & history

PHONX AI — OUTCOMES

Measurable impact

~1s	90%+	60%	∞
average response time	intent accuracy	call volume reduced	scale without headcount

WHAT CHANGED

- Insurance agencies scaled support capacity without growing their teams
- Policyholders receive consistent, compliant answers on every call
- Agents freed to handle complex edge cases requiring human judgment
- Phone operations transformed into a scalable infrastructure layer

Python · Go · Node.js · LLM Orchestration · Twilio · Proprietary STT/TTS · gRPC · Event Buses · Docker · K8s

ENTERPRISE VKG

Cross-Domain Compliance Engine for Research Teams

Industry: Pharma Research | Role: Lead Architect

5,000+
documents indexed

98%+
contextual accuracy

7
regulators unified

A hybrid retrieval system that stores semantic embeddings and graph relationships together — so AI retrieves not just matching text, but surrounding context, dependencies, and compliance logic needed for reliable decisions. Each regulation, SOP, and domain entity becomes a node in the graph with vector embeddings. Runs entirely on-prem within the client's secure private network with zero cloud dependency. Every decision is audit-logged and tamper-evident for regulatory inspection.

Ingestion Service
Parses PDF, Word, HTML across 7 sources

Embedding Service
Domain-tuned transformers

Neo4j + Vector Index
Graph DB with rule-based edges

Compliance Engine
Regulatory rules & conflict detection

Search & Validation API
GraphQL + REST contextual retrieval

Audit Log Layer
Tamper-evident logs for inspection

ENTERPRISE VKG — OUTCOMES

Measurable impact

5,000+
documents indexed

98%+
contextual accuracy

Days→Min
SOP validation time

7
regulators unified

WHAT CHANGED

- Research teams validate SOPs and resolve knowledge gaps in minutes instead of days
- Compliance officers have full audit-ready traceability for every query and response
- High-precision reasoning across interconnected regulatory data
- Teams move faster without increasing compliance risk

Python · FastAPI · Node.js · Transformer Models (domain-tuned) · Faiss · Neo4j · GraphQL · Docker · Kubernetes (on-prem)

SATELLITE ML

Geospatial ML for Large-Area Monitoring & Field Operations

Industry: Government / Law Enforcement | Role: ML Developer

0.87
F1 score

5 min
per 10×10 km

GIS
ready shapefiles

An end-to-end geospatial ML workflow developed in collaboration with Scanpoint Geomatics Limited as part of a government-linked initiative. Tiles large satellite scenes into overlapping geographic patches, classifies each with a custom CNN trained on the spectral signature of opium poppy fields, then reconstructs coordinate-precise outputs as GIS shapefiles. Post-processing merges tile predictions, suppresses noise artefacts, and produces standardised, auditable shapefile outputs ready for direct import into existing GIS workflows.

Ingestion Module
Large satellite files & geospatial metadata

Tiling Engine
Overlapping tiles with geographic index

Preprocessing Pipeline
Spectral normalisation class-balanced sampling

CNN Classifier
Custom arch for poppy spectral signature

Post-Processing
Merge predictions remove noise

GIS Output Layer
Shapefiles for QGIS & gov systems

SATELLITE ML — OUTCOMES

Measurable impact

0.87
F1 score on held-out test

5 min
per 10×10 km scene

GPS
coordinate-precise field outputs

↓↓
manual analyst hours reduced

WHAT CHANGED

- Field investigation teams received precise coordinates of suspicious cultivation areas
- Targeted operations enabled, dramatically reducing analyst workload
- Reproducible, Docker-based pipeline satisfied evidentiary standards for operational use
- Outputs plug directly into QGIS and government GIS systems without additional processing

Python · PyTorch · Custom CNN · OpenCV · GDAL · GeoPandas · QGIS · Docker

AVARA LABS

AI-Powered Email Automation for E-Commerce Support

Industry: E-commerce / Support | Role: Senior AI Engineer

~80%
queries automated

Multi
brand support

GDPR
compliant by design

When an email is received, the system ingests the message via platform connectors. The email content is parsed to extract message context — customer data, order history, and conversation thread. The intent classification framework determines the request type. If the intent is AI-safe, it passes into the AI response engine which retrieves conversation history, customer data, order history, and combines that with company policy to generate a reply. If the intent should not be handled by AI, the email is routed directly to a human agent.

Platform Connectors
Shopify, WooCommerce Magento integrations

Email Ingestion
Read, parse, extract message context

Intent Classification
Transformer model intent & urgency

Routing Engine
AI-safe vs human agent decision

AI Response Engine
Context + policy brand-safe replies

Human Agent Queue
Non-AI-safe intents to support staff

AVARA LABS — OUTCOMES

Measurable impact

~80%
queries automated end-to-end

Faster
response times across the board

Multi
brands supported simultaneously

GDPR
compliant by design

WHAT CHANGED

- Support teams freed from repetitive queries and redirected to complex, human-sensitive situations
- Brands saw meaningful improvements in response time and consistency
- Handled more volume with the same headcount while protecting brand voice
- Policy-grounded replies enforced across all supported platforms

Python · FastAPI · Transformer Intent Detection · Generative Response Layer · Shopify API · WooCommerce API · Magento API · PostgreSQL · Redis · Celery

KRISH SHAH

Need an AI system that feels sharp to clients and solid underneath?

Let's talk about the build, the constraints, and the version of this system that is actually worth shipping.

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